A MINI PROJECT ON

PDF Maker Application

SUBMITTED TO THE
PROJECT COORDINATOR FOR THE SUBJECT OF

ANDROID PROGRAMMING

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A.Y. 2025-2026

CERTIFICATE

THIS IS TO CERTIFY THE MINI PROJECT REPORT ENTITLED

"PDF Maker Application "

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Project Guide HOD

Place : Nashik

Date :

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1. Abstract

The **PDF Maker Application** is a software system developed to simplify the creation, editing, and management of PDF documents in a fast and user-friendly way. In today's digital world, PDF files are widely used for sharing documents because they preserve formatting, support security features, and are compatible across different devices and operating systems. This application is designed to help users generate professional PDF files from text, images, reports, forms, and other digital content.

The main objective of this project is to provide an easy platform where users can create PDF documents without requiring advanced technical knowledge. The system allows users to add text, insert images, format pages, merge multiple files, split documents, and save outputs in PDF format. Additional features such as password protection, page numbering, and document preview can also be included to enhance usability.

The application is developed using modern programming technologies such as Java, Python, PHP, or web-based frameworks depending on project requirements. It focuses on providing a clean interface, faster processing speed, and reliable performance. The system reduces manual effort and improves productivity for students, offices, businesses, and personal users.

This project demonstrates the practical implementation of document automation and digital file management. The **PDF Maker Application** is useful in educational institutions, corporate offices, and administrative sectors where frequent document generation is required. Future improvements may include cloud storage integration, digital signatures, and mobile application support.

Keywords:

- PDF Maker Application
- PDF Generation
- Document Management
- File Conversion
- Report Creation
- Digital Documentation
- User-Friendly Interface
- Data Processing
- Password Protection
- File Sharing
- Document Automation
- Cross-Platform Compatibility
- Cloud Integration
- Secure Documents
- Software Application

2. Introduction s Problem Definition

Introduction

The **PDF Maker Application** is a modern software solution designed to create, edit, convert, and manage PDF documents efficiently. In the digital era, Portable Document Format (PDF) files are commonly used for sharing official documents, reports, resumes, invoices, and academic files because they maintain the original layout across all devices and operating systems. Many users need a simple and reliable system to generate PDF files without using complex professional software.

This project aims to develop a user-friendly application that allows users to create PDF documents from text, images, and other file formats quickly. The application can include features such as document formatting, merging files, splitting pages, password protection, and preview options. It helps students, office workers, businesses, and general users save time and improve productivity.

The PDF Maker Application provides an easy interface and efficient processing system, making document creation simple and accessible. It also supports secure sharing of files and organized digital record management. This project represents an important step toward paperless work environments and smart document handling systems.

Problem Definition

Many users face difficulties while creating and managing PDF documents using traditional or expensive software tools. Existing applications may be complicated, costly, or require technical knowledge. Small businesses, students, and regular users often need a simple solution for converting files into PDF format and organizing documents easily.

Common problems faced by users include:

1. Difficulty in creating professional PDF documents quickly.
2. High cost of premium PDF editing software.
3. Lack of simple tools for merging, splitting, or converting files.
4. Risk of losing formatting during file sharing.
5. Limited security features in free applications.
6. Time-consuming manual document management processes.

The **PDF Maker Application** is developed to solve these problems by providing a low-cost, easy-to-use, and efficient platform for PDF creation and management.

3. Objectives:

1. **To develop a user-friendly web application** using HTML, CSS, and JavaScript for creating PDF files easily.
2. **To provide an attractive interface** with responsive design using CSS for better user experience on desktop and mobile devices.
3. **To enable users to enter text and generate PDF documents** directly through the web application.
4. **To allow image insertion** into PDF files for creating professional documents such as resumes, reports, and certificates.
5. **To implement real-time preview features** using JavaScript before downloading the PDF file.
6. **To reduce manual document creation time** by automating PDF generation.
7. **To provide download functionality** so users can save generated PDF files instantly.
8. **To create a lightweight and fast system** that works in any modern web browser without installing extra software.
9. **To improve document formatting options** such as font size, color, alignment, and page layout.
10. **To offer a free and accessible solution** for students, offices, and general users needing quick PDF creation.
11. **To enhance future scalability** by allowing integration of advanced features like password protection, cloud storage, and file merging.

4. Scope of the Project

The **PDF Maker Application** project has a wide scope in the field of digital document creation and management. It is designed to help users generate, edit, and download PDF files easily through a web-based platform developed using HTML, CSS, and JavaScript.

The **PDF Maker Application** is designed to provide an easy and efficient way to create PDF documents through a web-based platform. Developed using HTML, CSS, and JavaScript, the application allows users to generate PDF files directly from their browser without installing additional software. This makes the system simple, accessible, and convenient for all types of users.

The project scope includes creating documents with text, headings, paragraphs, and images in a professional format. Users can customize the appearance of documents by changing font styles, colors, alignment, and page layouts. The application also provides an instant download feature, allowing users to save their created files quickly in PDF format for future use or sharing.

The PDF Maker Application can be used in many areas such as education, business, offices, and personal work. Students can create assignments, resumes, and project reports, while offices can prepare invoices, notices, and official documents. It helps users save time, reduce manual effort, and improve productivity by automating the document creation process.

The future scope of the project is also wide, as advanced features can be added later. These include password protection, digital signatures, cloud storage, file merging, splitting, and online sharing options. The application also supports the concept of a paperless environment by promoting digital documentation and reducing paper usage.

1. Encourages digital documentation and reduces paper usage in organizations and institutions.

Overall, the scope of the **PDF Maker Application** is broad and practical, making it useful for personal, academic, and professional purposes.

From a technical perspective, the scope includes the use of:

- HTML
- CSS
- JavaScript
- PHP
- MySQL
- TensorFlow.js
- Teachable Machine
- GitHub
- InfinityFree Hosting
-

5. Requirements & Constraints

The PDF Maker Application requires a basic software and hardware environment for smooth development and operation. The system can be developed using HTML, CSS, and JavaScript as the main technologies. A modern web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge is required to run the application. For development, a code editor like Visual Studio Code can be used. The hardware requirements include a computer or laptop with at least 4 GB RAM, a dual-core processor, and sufficient storage space. The application can also run on mobile devices if designed responsively. A stable internet connection may be needed for online features or downloading libraries, although the core application can work offline if properly configured.

Some constraints of the project include limited advanced editing features compared to professional PDF software. Since the application is browser-based, performance may depend on the user's device speed and browser compatibility. Large files with many images may take more time to process or download. Security features such as password protection may require additional libraries or backend support.

Another limitation is that the project mainly focuses on basic PDF creation and formatting. Features such as digital signatures, cloud synchronization, OCR scanning, and collaborative editing may not be available in the initial version. However, these constraints can be reduced in future updates by integrating advanced technologies and server-side support.

Hardware Requirements

- Laptop/Desktop for development
- Smartphone for testing
- Minimum 4 GB RAM
- Internet connection
- Basic processor system

Software Requirements

- Visual Studio Code
- XAMPP Server
- PHP
- MySQL Database
- HTML
- CSS
- JavaScript
- Bootstrap
- TensorFlow.js

- Teachable Machine
- GitHub
- InfinityFree Hosting

6. System Architecture & Design

The PDF Maker Application follows a simple and efficient web-based architecture developed using HTML, CSS, and JavaScript. The system is mainly divided into three layers: the user interface layer, the processing layer, and the output layer. This structure helps in maintaining smooth interaction between the user and the application.

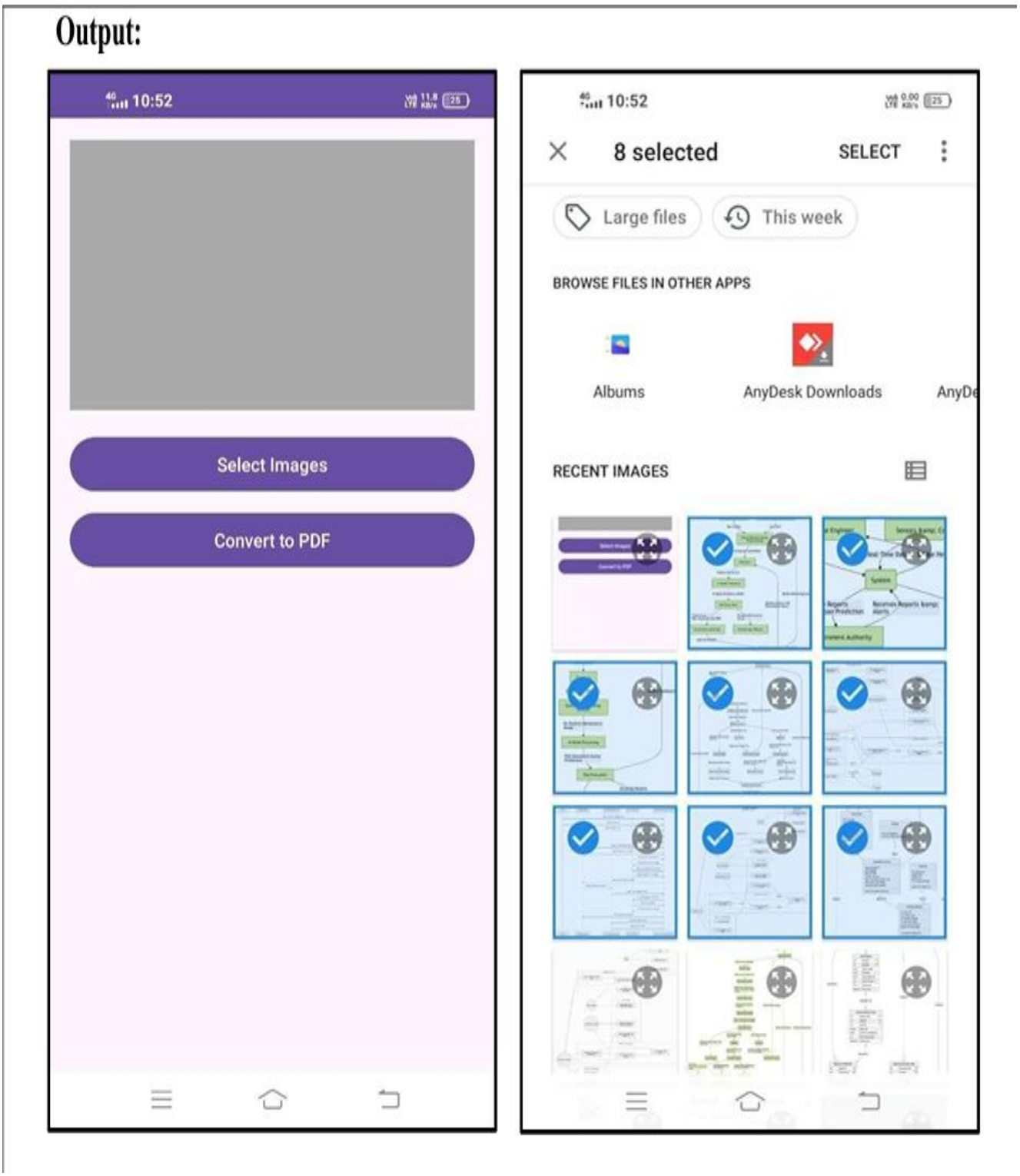
The User Interface Layer is created using HTML and CSS. It provides forms, buttons, text areas, image upload options, and formatting tools where users can enter content and customize their documents. CSS is used to design an attractive, responsive, and easy-to-use layout so that the application works properly on computers, tablets, and mobile devices.

The Processing Layer is handled by JavaScript. It collects the user input, applies formatting options, manages images, and processes the data into a structured document format. JavaScript libraries such as jsPDF can be used to convert the content into a PDF file. This layer controls the main functionality and ensures quick generation of documents.

The Output Layer is responsible for displaying a preview and allowing the user to download the generated PDF file. After processing, the final document is created and saved in PDF format on the user's device. The design of the system is modular, so future features such as password protection, cloud storage, and file merging can be added easily.

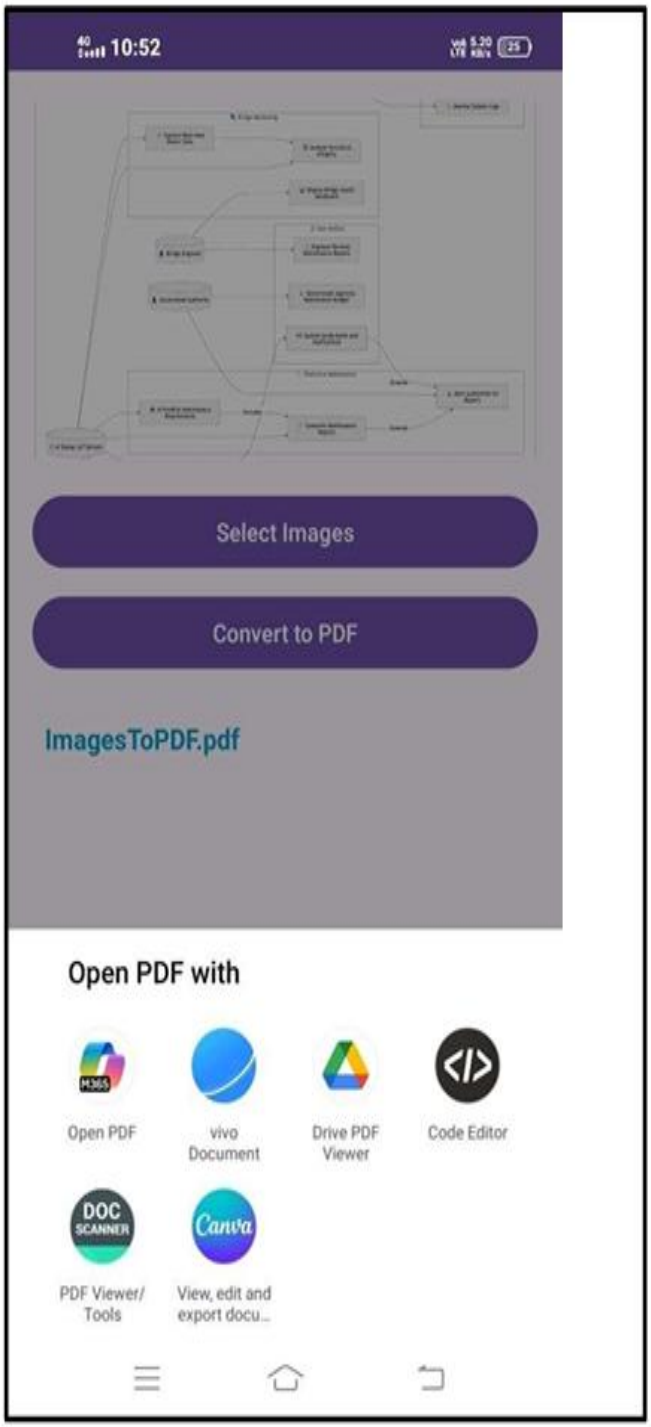
UI Interface :

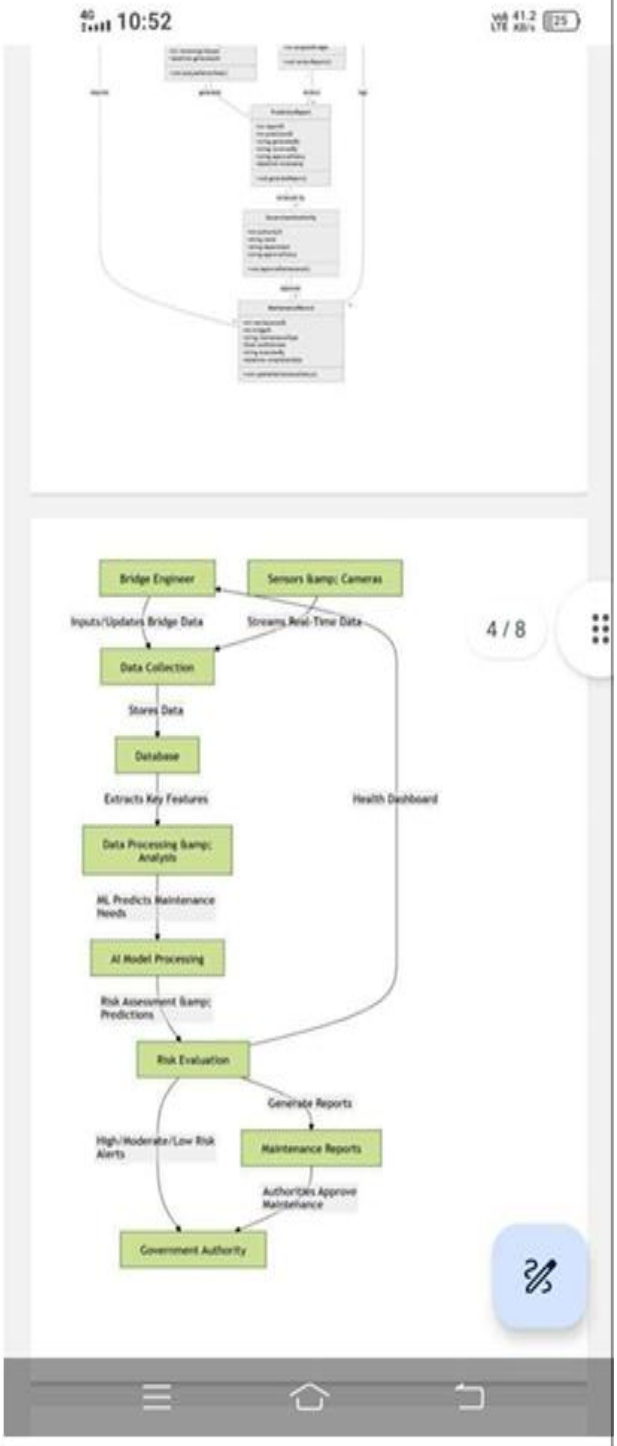
UI



10:52
Select Images
Convert to PDF

10:52
Select Images
Convert to PDF
ImagesToPDF.pdf
PDF saved in Documents folder


10:52
Select Images
Convert to PDF
ImagesToPDF.pdf
Open PDF with
Open PDF
vivo Document
Drive PDF Viewer
Code Editor
PDF Viewer/ Tools
View, edit and export docu...


10:52
4 / 8
Bridge Engineer
Sensors & Cameras
Inputs/Updates Bridge Data
Streams Real-Time Data
Data Collection
Stores Data
Database
Extracts Key Features
Health Dashboard
Data Processing & Analysis
ML Predicts Maintenance Needs
AI Model Processing
Risk Assessment & Predictions
Risk Evaluation
Generate Reports
High/Moderate/Low Risk Alerts
Maintenance Reports
Authorities Approve Maintenance
Government Authority

7. Implementation

The implementation of the PDF Maker Application is carried out using web technologies such as HTML, CSS, and JavaScript. HTML is used to create the structure of the application, including text fields, buttons, file upload options, and formatting controls. CSS is used to design the user interface, making the application attractive, responsive, and easy to use on different devices.

JavaScript is used to implement the core functionality of the system. It captures user input such as text, images, titles, and formatting choices. Event handling methods are used for button clicks, file selection, and preview generation. JavaScript also manages the logic required to organize content into a proper document format before converting it into a PDF file.

To generate the PDF document, a JavaScript library such as jsPDF or similar tools can be integrated. This library converts the entered content into a downloadable PDF file. Users can create documents, add images, customize font size or alignment, preview the result, and then download the final file directly to their device.

The application is tested in modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge to ensure compatibility and smooth performance. The modular implementation allows future improvements such as password protection, cloud storage, document merging, and advanced editing features.

7.1 UI INTERFACE CODE:

```html
<!DOCTYPE html>
<html>
<head>
    <title>Images to PDF</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<h1>Images to PDF Converter</h1>

<input type="file" id="imageInput" multiple accept="image/*">

<div id="preview"></div>

<button id="convertBtn">Convert to PDF</button>

<p id="status"></p>

<!-- jsPDF Library -->
<script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf/2.5.1

<script src="script.js"></script>

</body>
</html>
```

index.html

index.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Images to PDF</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<h1>Images to PDF Converter</h1>

<input type="file" id="imageInput" multiple accept="image/

<div id="preview"></div>

<button id="convertBtn">Convert to PDF</button>

<p id="status"></p>

<!-- jsPDF Library -->
<script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf/

<script src="script.js"></script>

</body>
</html>
```

Program:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="16dp">
<ImageView
android:id="@+id/imageView"
android:layout_width="match_parent"
android:layout_height="200dp"
android:scaleType="centerCrop"
android:background="@android:color/darker_gray" />
<Button
android:id="@+id/btnSelectImage"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Select Images"
android:layout_marginTop="16dp" />
<Button
android:id="@+id/btnConvertToPdf"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Convert to PDF"
android:layout_marginTop="8dp" />
<TextView
android:id="@+id/pdfNameTextView"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="PDF Name" android:textSize="16sp"
android:textStyle="bold"
android:padding="8dp"
android:visibility="gone"
android:textColor="@android:color/holo_blue_dark"
android:layout_marginTop="16dp" />
</LinearLayout>
```

MainActivity.java

```
package com.example.pdfmaker;
import android.content.ContentValues;
import android.content.Intent;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.net.Uri;
import android.os.Build;
import android.os.Bundle;
import android.provider.MediaStore;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
```

```
import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;
import com.itextpdf.text.Document;
import com.itextpdf.text.Image;
import com.itextpdf.text.pdf.PdfWriter;
import java.io.ByteArrayOutputStream;
import java.io.FileOutputStream;
import java.io.InputStream;
import java.io.OutputStream;
import java.util.ArrayList; public class MainActivity extends AppCompatActivity {
private ImageView imageView;
private TextView pdfNameTextView;
private final ArrayList<Uri> imageUris = new ArrayList<>();
private Uri pdfUri = null;
private final ActivityResultLauncher<Intent> pickImagesLauncher = registerForActivityResult(
new ActivityResultContracts.StartActivityForResult(),
result -> {
if (result.getResultCode() == RESULT_OK && result.getData() != null) {
imageUris.clear();
Intent data = result.getData();
if (data.getClipData() != null) { // Multiple images selected
int count = data.getClipData().getItemCount();
for (int i = 0; i < count; i++) {
Uri imageUri = data.getClipData().getItemAt(i).getUri();
imageUris.add(imageUri);
}
} else if (data.getData() != null) { // Single image selected
imageUris.add(data.getData());
}
// Display first selected image
if (!imageUris.isEmpty()) {
try {
InputStream inputStream = getContentResolver().openInputStream(imageUris.get(0));
Bitmap bitmap = BitmapFactory.decodeStream(inputStream);
imageView.setImageBitmap(bitmap);
} catch (Exception e) {
e.printStackTrace();
}
}
}
} );
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
imageView = findViewById(R.id.imageView);
pdfNameTextView = findViewById(R.id.pdfNameTextView);
Button btnSelectImages = findViewById(R.id.btnSelectImage);
Button btnConvertToPdf = findViewById(R.id.btnConvertToPdf);
btnSelectImages.setOnClickListener(v -> openGallery());
btnConvertToPdf.setOnClickListener(v -> {
if (!imageUris.isEmpty()) {
```

```
convertImagesToPdf();
} else {
Toast.makeText(this, "Please select images first", Toast.LENGTH_SHORT).show();
}
});
pdfNameTextView.setOnClickListener(v -> {
if (pdfUri != null) {
openPdfExternally();
} else {
Toast.makeText(this, "No PDF created yet!", Toast.LENGTH_SHORT).show();
}
});
}
private void openGallery() {
Intent intent = new Intent(Intent.ACTION_GET_CONTENT);
intent.setType("image/*");
intent.putExtra(Intent.EXTRA_ALLOW_MULTIPLE, true);
pickImagesLauncher.launch(Intent.createChooser(intent, "Select Images"));
} private void convertImagesToPdf() {
Document document = new Document();
OutputStream outputStream;
try {
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.Q) { // Android 10+
ContentValues values = new ContentValues();
values.put(MediaStore.MediaColumns.DISPLAY_NAME, "ImagesToPDF.pdf");
values.put(MediaStore.MediaColumns.MIME_TYPE, "application/pdf");
values.put(MediaStore.MediaColumns.RELATIVE_PATH, "Documents/");
pdfUri = getContentResolver().insert(MediaStore.Files.getContentUri("external"), values);
outputStream = getContentResolver().openOutputStream(pdfUri);
} else { // Android 9 and below
String filePath = getExternalFilesDir(null) + "/ImagesToPDF.pdf";
pdfUri = Uri.parse(filePath);
outputStream = new FileOutputStream(filePath);
}
PdfWriter.getInstance(document, outputStream);
document.open();
for (Uri imageUri : imageUris) {
InputStream inputStream = getContentResolver().openInputStream(imageUri);
Bitmap bitmap = BitmapFactory.decodeStream(inputStream);
ByteArrayOutputStream stream = new ByteArrayOutputStream();
bitmap.compress(Bitmap.CompressFormat.JPEG, 100, stream);
byte[] byteArray = stream.toByteArray();
Image image = Image.getInstance(byteArray);
image.scaleToFit(500, 700);
document.add(image);
document.newPage();
}
document.close();
outputStream.close(); pdfNameTextView.setText("ImagesToPDF.pdf");
pdfNameTextView.setVisibility(View.VISIBLE);
Toast.makeText(this, "PDF saved in Documents folder", Toast.LENGTH_LONG).show();
} catch (Exception e) {
e.printStackTrace();
```

```
Toast.makeText(this, "Error Creating PDF", Toast.LENGTH_SHORT).show();
}
}
private void openPdfExternally() {
Intent intent = new Intent(Intent.ACTION_VIEW);
intent.setDataAndType(pdfUri, "application/pdf");
intent.setFlags(Intent.FLAG_GRANT_READ_URI_PERMISSION);
startActivity(Intent.createChooser(intent, "Open PDF with"));
}
}
```

8. Verification & Validation:

Verification and validation are important processes used to ensure that the PDF Maker Application works correctly and meets user requirements. Verification focuses on checking whether the system is developed according to the design, coding standards, and project specifications. Validation ensures that the final application performs the expected tasks such as creating, formatting, and downloading PDF files successfully.

During verification, the source code developed using HTML, CSS, and JavaScript is reviewed and tested step by step. Each module such as text input, image upload, formatting tools, and download buttons is checked individually. Errors in layout, functionality, or coding logic are identified and corrected during this stage.

Validation is performed by testing the application with real user requirements. Users enter text, add images, apply formatting, and generate PDF documents to verify that the output matches expectations. The generated files are checked for correct layout, readable content, proper image placement, and successful downloading on different devices and browsers.

Different testing methods such as unit testing, integration testing, system testing, and user acceptance testing are used to improve reliability. The application is also tested in Google Chrome, Mozilla Firefox, and Microsoft Edge for browser compatibility. Proper verification and validation help ensure that the PDF Maker Application is stable, accurate, and user-friendly.

9. Results & Discussion:

The implementation of the **PDF Maker Application** produced successful results in creating a simple, efficient, and user-friendly platform for generating PDF documents. The application was able to accept user input such as text, headings, and images, and convert the content into properly formatted PDF files. Users could easily create and download documents without requiring any special software installation.

The application developed using HTML, CSS, and JavaScript showed smooth performance in modern web browsers. The user interface was responsive and easy to understand, allowing users to generate files quickly. Features such as font formatting, alignment, and image insertion improved the quality of the final PDF output.

During testing, the system successfully generated PDF files for different purposes such as resumes, reports, assignments, and official documents. The download process worked correctly, and the files maintained their layout across different devices. This demonstrated that the application is reliable for basic document creation tasks and can save time compared to manual methods.

The discussion of results shows that the project achieved its main objectives of simplicity, speed, and accessibility. Some limitations were observed, such as restricted advanced editing options and slower processing for large files with many images. However, the overall performance was satisfactory, and the system can be enhanced in the future with features like password protection, cloud storage, file merging, and digital signatures.

LearningOutcomesAchieved:

a) **InterpretfeaturesofAndroidoperatingsystem.**

b) ConfigureAndroidenvironmentanddevelopmenttools.

c)Developrichuserinterfacesbyusinglayoutsandcontrols

d)Useuserinterfacecomponentforandroidapplicationdev elopment.

e) CreateAndroidapplicationusingdatabase.

f) PublishAndroidapplications.

Conclusion & Lesson Learned

The PDF Maker Application was successfully designed and implemented as a web-based solution for creating PDF documents quickly and efficiently. The project achieved its main objective of providing a simple, user-friendly, and low-cost platform where users can generate professional PDF files without using complex software. By using HTML, CSS, and JavaScript, the application delivered smooth performance, responsive design, and useful features such as text formatting, image insertion, preview, and download options.

The project proved that web technologies can be effectively used to build practical document management applications. The generated PDF files were accurate, readable, and compatible across multiple devices and browsers. This system can be useful for students, offices, businesses, and personal users who regularly need reports, resumes, assignments, invoices, and official documents in PDF format.

During the development process, several valuable lessons were learned. Proper planning and system design are important for successful project implementation. The use of JavaScript libraries such as jsPDF simplified the PDF generation process and reduced development time. Testing across different

browsers helped improve compatibility and user experience. The project also highlighted the importance of responsive design, clean coding practices, and error handling.

Overall, the project was a successful learning experience in front-end web development, document automation, and software problem-solving. Future improvements such as password protection, cloud storage, digital signatures, and advanced editing tools can further enhance the application.

Appendix & References

Appendix

The appendix section contains supporting materials related to the PDF Maker Application project. It includes sample input forms, screenshots of the user interface, source code modules, testing reports, and generated PDF output examples. These materials help in understanding the practical implementation of the project.

Appendix A – Tools and Technologies Used

- **HTML – Used for creating the structure of web pages**
- **CSS – Used for styling and responsive design**
- **JavaScript – Used for application logic and interactivity**
- **jsPDF – Used for generating PDF files**
- **Visual Studio Code – Used for coding and development**

Appendix B – Hardware Requirements

- **Computer/Laptop with minimum 4 GB RAM**
- **Dual-Core Processor or higher**
- **500 MB free storage space**
- **Internet connection (optional for updates/libraries)**

Appendix C – Software Requirements

- **Google Chrome**
- **Mozilla Firefox**

- **Microsoft Edge**
- **Windows 10 or higher**

References

1. **Mozilla Developer Network (MDN) Web Docs – HTML, CSS, JavaScript Documentation**
2. **jsPDF Official Documentation**
3. **W3Schools – Web Development Tutorials**
4. **Microsoft – Visual Studio Code Documentation**
5. **Standard Web Development Books and Online Resources on PDF Generation Technologies**

References

1. **Mozilla Developer Network.** ***HTML, CSS, and JavaScript Documentation.***
2. **jsPDF.** ***Official Documentation for PDF Generation using JavaScript.***
3. **W3Schools.** ***Tutorials on HTML, CSS, JavaScript.***
4. **Microsoft.** ***Visual Studio Code User Guide and Documentation.***
5. **Google Chrome.** ***Browser Testing and Developer Tools.***
6. **Mozilla Firefox.** ***Browser Compatibility Testing Tools.***
7. **Standard textbooks and online resources related to Web Development and PDF Processing Systems.**